# Supplementary Material
# How Hard Is the Attack? A Complexity-Theoretic Systematization of Machine Learning Security

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## 1. Classifying the corpus

This section reports the corpus in two layers. Tables 2 to 4 cross-tabulate features against one another, with works in the cells by citation number. Tables 5 and 6 give one row per work, so that every cell above is traceable to the paper behind it. Every value comes from a fixed vocabulary settled before the pass began.

The cross-tabulations are the layer that carries an argument, because in them an empty cell is a combination nobody has published. The value `none` is an answer and not a missing entry, recording that a work constrains no party at all. Of the 199 works classified so far, 33 state no resource bound on any party, and 0 fall outside the frame in the sense that deleting every resource restriction leaves the main claim unchanged.

*This is a partial pass. 199 of the corpus's works are classified. The remainder are not yet placed.*

TABLE 1: The corpus by pipeline phase (rows) against access level (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies.

| **phase** | **hard label** | **scores** | **black box** | **white box** | **weights** | **gradients** | **channel** | **none** | ***unassigned*** | Σ |
|---|---|---|---|---|---|---|---|---|---|---|
| train time | [62] | | [105], [146] | [68], [71], [137], [140], [166], [185], [195], [197] | | [28] | [192] | [4], [12], [17], [22], [34], [36], [46], [70], [72], [80], [131], [160], [162], [168], [190], [191] | | 29 |
| test time | [92] | | [32], [57], [96], [97], [101], [158], [167], [174] | [6], [13], [49], [50], [55], [65], [66], [98], [143], [144], [173], [176], [177], [181], [184], [187] | | | [114] | [10], [40], [43], [51], [78], [85], [93], [108], [136], [142], [150], [151], [156], [170], [199] | | 41 |
| post hoc | | [16], [53], [54], [61], [86], [117], [122], [159], [172], [180], [196] | [52], [74], [77], [109], [124], [152], [171], [188] | [25], [58], [59], [95], [100], [126], [130], [134], [145], [157], [163], [182], [186], [193] | [121] | [127] | | [63], [90], [149], [153], [179] | [164] | 41 |
| n/a | | [79] | | | | | | [1], [2], [3], [5], [7], [8], [9], [11], [14], [15], [18], [19], [20], [21], [23], [24], [26], [27], [29], [30], [31], [33], [35], [37], [38], [39], [41], [42], [44], [45], [47], [48], [56], [60], [64], [67], [69], [73], [75], [76], [81], [82], [83], [84], [87], [88], [89], [91], [94], [99], [102], [103], [104], [106], [107], [110], [111], [112], [113], [115], [116], [118], [119], [120], [123], [125], [128], [129], [132], [133], [135], [138], [139], [141], [147], [148], [154], [155], [161], [165], [169], [175], [178], [183], [189], [194], [198] | | 88 |
| Σ | 2 | 12 | 18 | 38 | 1 | 2 | 2 | 123 | 1 | 199 |

TABLE 2: The corpus by constrained party (rows) against resource unit (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies. A dagger marks works where the resource appears only as a side condition rather than as a parameter the work assesses on purpose.

| party | time | circuit size | queries | samples | memory | algo class | parameters | perturbation | corruption | rounds | privacy budget | circuit depth | success p |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| attacker | [11], [26], [33], [45], [62], [63], [71], [74], [79], [81], [92], [100], [106], [117], [118], [126], [138], [147], [148], [180], [185], [192], [28], [54], [95], [174], [195], [197]† | [1], [23], [38], [48], [99], [101] | [5], [42], [52], [57], [122], [145], [53], [61], [96], [152], [172]† | [121], [164]† | [24], [83], [84], [110] | [21], [60], [89], [124], [189] | | [6], [13], [50], [65], [66], [78], [93], [108], [114], [136], [142], [150], [151], [156], [170], [173], [176], [177], [181], [184], [187], [49], [55]† | [137], [140], [146], [160], [168] | | | [107] | |
| learner | [15], [19], [20], [31], [34], [36], [37], [40], [44], [47], [69], [72], [76], [80], [119], [123], [135], [141], [165], [178], [198] | [103] | [22]† | [7], [10], [29], [51], [70], [73], [75], [82], [88], [111], [112], [125], [128], [139], [143], [144], [183], [190] | [162] | [2], [3], [8], [17], [43], [56], [67], [91], [94], [104], [131], [175], [191] | [46], [102], [115] | | | | | | |
| auditor | [58], [68], [105], [109], [130], [32], [196]† | | | [153], [199] | | [35], [77], [116] | | | | | | | |
| mechanism | [27], [161] | | | | [87], [133] | [194]† | | | | | [4], [163]† | | |
| none | | | | | | | | | | | | | |
| Σ | 58 | 7 | 12 | 22 | 7 | 22 | 3 | 23 | 5 | 0 | 2 | 1 | 0 |

TABLE 3: The corpus by resource unit (rows) against tightness of the bound (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies.

| **unit** | **unstated** | **poly** | **quasi-poly** | **class** | **budget** | **conditional** | **exponent** | **measured** | Σ |
|---|---|---|---|---|---|---|---|---|---|
| time | [126], [196] | [27], [37], [40], [58], [62], [63], [68], [72], [79], [80], [81], [92], [100], [105], [106], [109], [117], [118], [119], [123], [130], [141], [147], [148], [178], [180], [185], [192], [198] | [34] | | [161], [197] | [11], [15], [20], [36], [44], [69] | [26], [32], [33], [45], [47], [76], [135], [138], [195] | [19], [28], [31], [54], [71], [74], [95], [165], [174] | 58 |
| circuit size | | [101] | [1], [48], [99] | | [38] | [103] | [23] | | 7 |
| queries | | | | | [22], [52], [96], [145], [152] | | [5], [42], [57], [122] | [53], [61], [172] | 12 |
| samples | | | | [70], [190] | [10], [51], [143], [144], [199] | | [7], [29], [73], [75], [82], [88], [111], [112], [121], [125], [128], [139], [153], [183] | [164] | 22 |
| memory | | | | | [162] | [87] | [24], [83], [84], [110], [133] | | 7 |
| algo class | | | | [2], [3], [8], [17], [21], [35], [43], [56], [60], [67], [77], [89], [91], [94], [104], [116], [124], [131], [175], [189], [191], [194] | | | | | 22 |
| parameters | | | | | [46], [102], [115] | | | | 3 |
| perturbation | | | | | [6], [13], [49], [50], [55], [65], [66], [78], [93], [108], [114], [136], [142], [150], [151], [156], [170], [173], [176], [177], [181], [184], [187] | | | | 23 |
| corruption | | | | | [137], [140], [146], [168] | | | [160] | 5 |
| rounds | | | | | | | | | 0 |
| privacy budget | | | | | [4], [163] | | | | 2 |
| circuit depth | | | | | | | [107] | | 1 |
| success prob | | | | | | | | | 0 |
| flops | | | | | | | [18], [113], [132], [155] | | 4 |
| none | [9], [12], [14], [16], [25], [30], [39], [41], [59], [64], [85], [86], [90], [97], [98], [120], [127], [129], [134], [149], [154], [157], [158], [159], [166], [167], [169], [171], [179], [182], [186], [188], [193] | | | | | | | | 33 |
| Σ | 35 | 30 | 4 | 24 | 46 | 8 | 38 | 14 | 199 |

TABLE 4: The corpus by attack family (rows) against constrained party (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies.

| **family** | **attacker** | **learner** | **auditor** | **mechanism** | **none** | Σ |
|---|---|---|---|---|---|---|
| membership | [54], [100], [121], [172], [185] | | [196] | | [16], [86], [90], [149], [159], [193] | 12 |
| attribute | | | | | | 0 |
| reconstruct | [28], [63], [79], [117], [122], [145] | | | | [25], [59], [134], [188] | 10 |
| distribution | [180], [192] | | | | | 2 |
| extract | [52], [53], [61], [74], [124] | | | | [171] | 6 |
| invert | | | | | | 0 |
| forge | [71], [84], [95], [126], [197] | | | | [127], [166], [182] | 8 |
| evade | [6], [13], [49], [50], [55], [65], [66], [78], [92], [93], [96], [101], [108], [114], [136], [142], [150], [151], [156], [170], [173], [174], [176], [177], [181], [184], [187] | [10], [40], [43], [51], [80], [143], [144] | [32] | | [97], [98] | 37 |
| evade audit | [57], [195] | | [153] | | [158], [179] | 5 |
| inject | | | [199] | | | 1 |
| poison | [137], [140], [146], [168] | [22], [70], [72], [190] | | | | 8 |
| backdoor | [152], [160] | | [58], [68], [105], [109], [130] | | [12], [157], [186] | 10 |
| multiple | | | | | [129], [154] | 2 |
| none | [1], [5], [11], [21], [23], [24], [26], [33], [38], [42], [45], [48], [60], [62], [81], [83], [89], [99], [106], [107], [110], [118], [138], [147], [148], [189] | [2], [3], [7], [8], [15], [17], [18], [19], [20], [29], [31], [34], [36], [37], [44], [46], [47], [56], [67], [69], [73], [75], [76], [82], [88], [91], [94], [102], [103], [104], [111], [112], [113], [115], [119], [123], [125], [128], [131], [132], [135], [139], [141], [155], [162], [165], [175], [178], [183], [191], [198] | [35], [77], [116] | [4], [27], [87], [133], [161], [194] | [9], [14], [30], [39], [41], [64], [120], [167], [169] | 95 |
| *unassigned* | [164] | | | [163] | [85] | 3 |
| Σ | 85 | 62 | 12 | 7 | 33 | 199 |

TABLE 5: Per-work reference layer: attack scenario and threat model.

| # | Target | Family | Phase | Access | Write | Frame |
|---|---|---|---|---|---|---|
| [90] | train data | membership | post hoc | none | read only | none |
| [159] | train data | membership | post hoc | scores | read only | none |
| [86] | train data | membership | post hoc | scores | read only | none |
| [185] | train data | membership | train time | white box | corpus | compute |
| [121] | train data | membership | post hoc | weights | read only | statistical |
| [172] | train data | membership | post hoc | scores | read only | compute |
| [196] | train data | membership | post hoc | scores | read only | compute |
| [54] | train data | membership | post hoc | scores | read only | compute |
| [16] | train data | membership | post hoc | scores | read only | none |
| [193] | train data | membership | post hoc | white box | read only | none |
| [149] | train data | membership | post hoc | none | none | none |
| [100] | train data | membership | post hoc | white box | read only | compute |
| [59] | train data | reconstruct | post hoc | white box | read only | none |
| [63] | train data | reconstruct | post hoc | none | read only | compute |
| [79] | train data | reconstruct | n/a | scores | read only | compute |
| [134] | train data | reconstruct | post hoc | white box | read only | none |
| [28] | train data | reconstruct | train time | gradients | read only | compute |
| [122] | train data | reconstruct | post hoc | scores | read only | compute |
| [145] | train data | reconstruct | post hoc | white box | read only | compute |
| [25] | train data | reconstruct | post hoc | white box | read only | none |
| [188] | train data | reconstruct | post hoc | black box | read only | none |
| [117] | train data | reconstruct | post hoc | scores | read only | compute |
| [180] | train data | distribution | post hoc | scores | read only | compute |
| [192] | train data | distribution | train time | channel | corpus | compute |
| [52] | parameters | extract | post hoc | black box | read only | compute |
| [171] | parameters | extract | post hoc | black box | read only | none |
| [74] | parameters | extract | post hoc | black box | read only | compute |
| [61] | parameters | extract | post hoc | scores | read only | compute |
| [124] | parameters | extract | post hoc | black box | read only | compute |
| [53] | parameters | extract | post hoc | scores | read only | compute |
| [182] | training run | forge | post hoc | white box | corpus | none |
| [84] | none | forge | n/a | none | none | compute |
| [127] | train data | forge | post hoc | gradients | corpus | none |
| [197] | training run | forge | train time | white box | model | compute |
| [95] | training run | forge | post hoc | white box | corpus | compute |
| [71] | training run | forge | train time | white box | model | compute |
| [126] | training run | forge | post hoc | white box | model | compute |
| [166] | training run | forge | train time | white box | corpus | none |
| [181] | outputs | evade | test time | white box | read only | capability |
| [55] | outputs | evade | test time | white box | read only | capability |
| [170] | outputs | evade | test time | none | read only | capability |
| [43] | outputs | evade | test time | none | read only | compute |
| [40] | outputs | evade | test time | none | none | compute |
| [108] | outputs | evade | test time | none | read only | capability |
| [93] | outputs | evade | test time | none | read only | capability |
| [10] | outputs | evade | test time | none | read only | statistical |
| [136] | outputs | evade | test time | none | read only | capability |
| [65] | outputs | evade | test time | white box | read only | capability |
| [151] | outputs | evade | test time | none | read only | capability |
| [92] | outputs | evade | test time | hard label | read only | compute |
| [101] | outputs | evade | test time | black box | read only | compute |
| [142] | outputs | evade | test time | none | read only | capability |
| [156] | outputs | evade | test time | none | read only | capability |
| [150] | outputs | evade | test time | none | read only | capability |
| [32] | outputs | evade | test time | black box | read only | compute |
| [98] | outputs | evade | test time | white box | read only | none |
| [97] | outputs | evade | test time | black box | read only | none |
| [78] | outputs | evade | test time | none | read only | capability |

*Table 5, continued*

| # | Target | Family | Phase | Access | Write | Frame |
|---|---|---|---|---|---|---|
| [66] | outputs | evade | test time | white box | read only | capability |
| [6] | outputs | evade | test time | white box | read only | capability |
| [49] | outputs | evade | test time | white box | read only | capability |
| [51] | outputs | evade | test time | none | read only | statistical |
| [187] | outputs | evade | test time | white box | read only | capability |
| [177] | outputs | evade | test time | white box | read only | capability |
| [173] | outputs | evade | test time | white box | read only | capability |
| [144] | outputs | evade | test time | white box | read only | statistical |
| [50] | outputs | evade | test time | white box | read only | capability |
| [184] | outputs | evade | test time | white box | read only | capability |
| [114] | outputs | evade | test time | channel | channel | capability |
| [176] | outputs | evade | test time | white box | read only | capability |
| [96] | outputs | evade | test time | black box | read only | compute |
| [174] | outputs | evade | test time | black box | read only | compute |
| [13] | outputs | evade | test time | white box | read only | capability |
| [143] | outputs | evade | test time | white box | none | statistical |
| [80] | none | evade | train time | none | none | compute |
| [179] | training run | evade audit | post hoc | none | none | none |
| [153] | training run | evade audit | post hoc | none | responses | statistical |
| [158] | outputs | evade audit | test time | black box | read only | none |
| [57] | outputs | evade audit | test time | black box | responses | compute |
| [195] | training run | evade audit | train time | white box | model | compute |
| [199] | outputs | inject | test time | none | channel | statistical |
| [168] | train data | poison | train time | none | corpus | capability |
| [140] | train data | poison | train time | white box | corpus | capability |
| [137] | train data | poison | train time | white box | corpus | capability |
| [146] | train data | poison | train time | black box | corpus | capability |
| [72] | train data | poison | train time | none | corpus | compute |
| [70] | train data | poison | train time | none | corpus | compute |
| [190] | train data | poison | train time | none | corpus | compute |
| [22] | train data | poison | train time | none | corpus | compute |
| [58] | parameters | backdoor | post hoc | white box | read only | compute |
| [157] | parameters | backdoor | post hoc | white box | read only | none |
| [152] | outputs | backdoor | post hoc | black box | read only | compute |
| [186] | parameters | backdoor | post hoc | white box | read only | none |
| [130] | parameters | backdoor | post hoc | white box | model | compute |
| [109] | parameters | backdoor | post hoc | black box | read only | compute |
| [105] | parameters | backdoor | train time | black box | model | compute |
| [12] | train data | backdoor | train time | none | corpus | none |
| [68] | parameters | backdoor | train time | white box | model | compute |
| [160] | parameters | backdoor | train time | none | corpus | capability |
| [129] | none | multiple | n/a | none | none | none |
| [154] | none | multiple | n/a | none | none | none |
| [133] | none | none | n/a | none | none | compute |
| [112] | none | none | n/a | none | none | compute |
| [9] | none | none | n/a | none | none | none |
| [45] | none | none | n/a | none | none | compute |
| [76] | none | none | n/a | none | none | compute |
| [17] | none | none | train time | none | none | compute |
| [35] | none | none | n/a | none | none | compute |
| [38] | none | none | n/a | none | none | compute |
| [110] | none | none | n/a | none | none | compute |
| [21] | none | none | n/a | none | none | compute |
| [44] | none | none | n/a | none | none | compute |
| [24] | none | none | n/a | none | none | compute |
| [60] | train data | none | n/a | none | none | compute |
| [118] | none | none | n/a | none | none | compute |
| [148] | none | none | n/a | none | none | compute |
| [113] | none | none | n/a | none | none | compute |

*Table 5, continued*

| # | Target | Family | Phase | Access | Write | Frame |
|---|---|---|---|---|---|---|
| [30] | none | none | n/a | none | none | none |
| [147] | train data | none | n/a | none | none | compute |
| [155] | none | none | n/a | none | none | compute |
| [62] | training run | none | train time | hard label | read only | compute |
| [1] | none | none | n/a | none | none | compute |
| [165] | none | none | n/a | none | none | compute |
| [4] | train data | none | train time | none | none | capability |
| [37] | none | none | n/a | none | none | compute |
| [69] | none | none | n/a | none | none | compute |
| [47] | none | none | n/a | none | none | compute |
| [14] | none | none | n/a | none | none | none |
| [87] | none | none | n/a | none | none | compute |
| [161] | train data | none | n/a | none | none | compute |
| [18] | none | none | n/a | none | none | compute |
| [29] | none | none | n/a | none | none | compute |
| [19] | none | none | n/a | none | none | compute |
| [94] | none | none | n/a | none | none | compute |
| [89] | none | none | n/a | none | none | compute |
| [36] | none | none | train time | none | none | compute |
| [15] | none | none | n/a | none | none | compute |
| [5] | none | none | n/a | none | none | compute |
| [26] | none | none | n/a | none | none | compute |
| [23] | none | none | n/a | none | none | compute |
| [11] | none | none | n/a | none | none | compute |
| [167] | outputs | none | test time | black box | corpus | none |
| [128] | none | none | n/a | none | none | compute |
| [103] | none | none | n/a | none | none | compute |
| [104] | none | none | n/a | none | none | compute |
| [82] | none | none | n/a | none | none | compute |
| [75] | none | none | n/a | none | none | compute |
| [141] | none | none | n/a | none | none | compute |
| [178] | none | none | n/a | none | none | compute |
| [31] | none | none | n/a | none | none | compute |
| [120] | none | none | n/a | none | none | none |
| [123] | none | none | n/a | none | none | compute |
| [194] | none | none | n/a | none | none | compute |
| [8] | none | none | n/a | none | none | compute |
| [77] | outputs | none | post hoc | black box | read only | compute |
| [39] | none | none | n/a | none | none | none |
| [91] | none | none | n/a | none | none | compute |
| [169] | none | none | n/a | none | none | none |
| [3] | none | none | n/a | none | none | compute |
| [67] | none | none | n/a | none | none | compute |
| [119] | none | none | n/a | none | none | compute |
| [183] | none | none | n/a | none | none | compute |
| [106] | none | none | n/a | none | none | compute |
| [56] | none | none | n/a | none | none | compute |
| [175] | none | none | n/a | none | none | compute |
| [191] | none | none | train time | none | none | compute |
| [64] | none | none | n/a | none | none | none |
| [2] | none | none | n/a | none | none | compute |
| [42] | none | none | n/a | none | none | compute |
| [189] | none | none | n/a | none | none | compute |
| [73] | none | none | n/a | none | none | compute |
| [102] | none | none | n/a | none | none | compute |
| [33] | none | none | n/a | none | none | compute |
| [138] | none | none | n/a | none | none | compute |
| [48] | none | none | n/a | none | none | compute |
| [99] | none | none | n/a | none | none | compute |

*Table 5, continued*

| # | Target | Family | Phase | Access | Write | Frame |
|---|---|---|---|---|---|---|
| [135] | none | none | n/a | none | none | compute |
| [34] | none | none | train time | none | none | compute |
| [162] | none | none | train time | none | none | compute |
| [20] | none | none | n/a | none | none | compute |
| [7] | none | none | n/a | none | none | compute |
| [88] | none | none | n/a | none | none | compute |
| [116] | none | none | n/a | none | none | compute |
| [132] | none | none | n/a | none | none | compute |
| [27] | none | none | n/a | none | none | compute |
| [107] | none | none | n/a | none | none | compute |
| [139] | none | none | n/a | none | none | compute |
| [111] | none | none | n/a | none | none | compute |
| [125] | none | none | n/a | none | none | compute |
| [81] | none | none | n/a | none | none | compute |
| [83] | none | none | n/a | none | none | compute |
| [115] | none | none | n/a | none | none | compute |
| [46] | none | none | train time | none | none | compute |
| [131] | train data | none | train time | none | none | compute |
| [198] | none | none | n/a | none | none | compute |
| [41] | none | none | n/a | none | none | none |
| [163] | train data | · | post hoc | white box | read only | capability |
| [85] | outputs | · | test time | none | responses | none |
| [164] | train data | · | post hoc | · | read only | statistical |

TABLE 6: Per-work reference layer: adversary assumptions and claim shape. Same rows and same order as Table 5.

| # | Party | Unit | Tightness | Baseline | 2nd party | 2nd notion | Assessed | Quantifier | Antecedent |
|---|---|---|---|---|---|---|---|---|---|
| [90] | none | none | unstated | stated | none | none | · | empirical | stated |
| [159] | none | none | unstated | stated | none | none | · | empirical | stated |
| [86] | none | none | unstated | stated | none | none | · | empirical | stated |
| [185] | attacker | time | poly | stated | none | none | explicit | conditional | stated |
| [121] | attacker | samples | exponent | stated | samples | · | explicit | universal | stated |
| [172] | attacker | queries | measured | stated | none | none | implicit | empirical | stated |
| [196] | auditor | time | unstated | stated | none | none | implicit | conditional | stated |
| [54] | attacker | time | measured | stated | none | none | implicit | empirical | stated |
| [16] | none | none | unstated | stated | none | none | · | empirical | stated |
| [193] | none | none | unstated | stated | none | none | · | conditional | stated |
| [149] | none | none | unstated | stated | none | none | · | empirical | stated |
| [100] | attacker | time | poly | stated | none | queries | explicit | conditional | stated |
| [59] | none | none | unstated | stated | none | · | · | universal | stated |
| [63] | attacker | time | poly | stated | time | none | explicit | conditional | stated |
| [79] | attacker | time | poly | absent | perturbation | queries | explicit | universal | stated |
| [134] | none | none | unstated | stated | none | · | · | existential | stated |
| [28] | attacker | time | measured | absent | none | none | implicit | empirical | stated |
| [122] | attacker | queries | exponent | stated | samples | none | explicit | conditional | stated |
| [145] | attacker | queries | budget | absent | privacy budget | none | explicit | universal | stated |
| [25] | none | none | unstated | stated | none | none | · | conditional | stated |
| [188] | none | none | unstated | n/a | none | none | · | empirical | stated |
| [117] | attacker | time | poly | stated | none | queries | explicit | conditional | stated |
| [180] | attacker | time | poly | stated | · | none | explicit | conditional | stated |
| [192] | attacker | time | poly | stated | time | none | explicit | conditional | stated |
| [52] | attacker | queries | budget | absent | none | none | explicit | conditional | stated |
| [171] | none | none | unstated | absent | none | none | · | empirical | stated |
| [74] | attacker | time | measured | stated | none | none | explicit | empirical | stated |
| [61] | attacker | queries | measured | absent | none | none | implicit | empirical | stated |
| [124] | attacker | algo class | class | absent | none | none | explicit | existential | stated |
| [53] | attacker | queries | measured | stated | none | time | implicit | empirical | stated |
| [182] | none | none | unstated | n/a | none | none | · | existential | stated |
| [84] | attacker | memory | exponent | stated | memory | none | explicit | universal | stated |
| [127] | none | none | unstated | absent | none | none | · | empirical | stated |
| [197] | attacker | time | budget | n/a | time | none | implicit | conditional | stated |
| [95] | attacker | time | measured | stated | none | none | implicit | empirical | stated |
| [71] | attacker | time | measured | stated | none | none | explicit | empirical | stated |
| [126] | attacker | time | unstated | absent | none | none | explicit | conditional | stated |
| [166] | none | none | unstated | n/a | none | none | · | empirical | stated |
| [181] | attacker | perturbation | budget | stated | none | none | explicit | existential | none |
| [55] | attacker | perturbation | budget | n/a | none | none | implicit | empirical | none |
| [170] | attacker | perturbation | budget | n/a | none | none | explicit | conditional | stated |
| [43] | learner | algo class | class | n/a | perturbation | · | explicit | existential | stated |
| [40] | learner | time | poly | stated | perturbation | none | explicit | conditional | stated |
| [108] | attacker | perturbation | budget | n/a | none | none | explicit | existential | stated |
| [93] | attacker | perturbation | budget | n/a | none | none | explicit | conditional | stated |
| [10] | learner | samples | budget | n/a | none | · | explicit | universal | stated |
| [136] | attacker | perturbation | budget | n/a | none | · | explicit | universal | stated |
| [65] | attacker | perturbation | budget | n/a | none | none | explicit | universal | stated |
| [151] | attacker | perturbation | budget | n/a | none | none | explicit | conditional | stated |
| [92] | attacker | time | poly | n/a | none | none | explicit | universal | stated |
| [101] | attacker | circuit size | poly | n/a | none | none | explicit | existential | stated |
| [142] | attacker | perturbation | budget | n/a | none | none | explicit | conditional | stated |
| [156] | attacker | perturbation | budget | n/a | none | none | explicit | conditional | stated |
| [150] | attacker | perturbation | budget | n/a | none | none | explicit | empirical | stated |
| [32] | auditor | time | exponent | stated | none | · | implicit | existential | stated |
| [98] | none | none | unstated | stated | none | · | · | conditional | stated |
| [97] | none | none | unstated | n/a | none | none | · | empirical | stated |

*Table 6, continued*

| # | Party | Unit | Tightness | Baseline | 2nd party | 2nd notion | Assessed | Quantifier | Antecedent |
|---|---|---|---|---|---|---|---|---|---|
| [78] | attacker | perturbation | budget | n/a | samples | none | explicit | conditional | stated |
| [66] | attacker | perturbation | budget | n/a | none | none | explicit | conditional | stated |
| [6] | attacker | perturbation | budget | stated | none | none | explicit | existential | none |
| [49] | attacker | perturbation | budget | n/a | none | none | implicit | empirical | none |
| [51] | learner | samples | budget | n/a | none | none | explicit | universal | stated |
| [187] | attacker | perturbation | budget | n/a | none | none | explicit | universal | stated |
| [177] | attacker | perturbation | budget | n/a | none | none | explicit | empirical | stated |
| [173] | attacker | perturbation | budget | n/a | samples | none | explicit | existential | stated |
| [144] | learner | samples | budget | stated | none | none | explicit | universal | stated |
| [50] | attacker | perturbation | budget | stated | none | time | explicit | empirical | stated |
| [184] | attacker | perturbation | budget | stated | none | none | explicit | existential | stated |
| [114] | attacker | perturbation | budget | n/a | samples | none | explicit | conditional | stated |
| [176] | attacker | perturbation | budget | n/a | samples | none | explicit | conditional | stated |
| [96] | attacker | queries | budget | stated | none | algo class | implicit | existential | none |
| [174] | attacker | time | measured | n/a | none | none | implicit | empirical | none |
| [13] | attacker | perturbation | budget | n/a | samples | none | explicit | universal | stated |
| [143] | learner | samples | budget | n/a | none | none | explicit | universal | stated |
| [80] | learner | time | poly | n/a | perturbation | none | explicit | conditional | stated |
| [179] | none | none | unstated | n/a | none | · | · | conditional | stated |
| [153] | auditor | samples | exponent | n/a | none | none | explicit | conditional | stated |
| [158] | none | none | unstated | n/a | none | none | · | existential | stated |
| [57] | attacker | queries | exponent | stated | none | time | explicit | existential | stated |
| [195] | attacker | time | exponent | n/a | none | · | implicit | existential | stated |
| [199] | auditor | samples | budget | n/a | none | none | explicit | conditional | stated |
| [168] | attacker | corruption | budget | stated | algo class | none | explicit | conditional | stated |
| [140] | attacker | corruption | budget | n/a | none | none | explicit | conditional | stated |
| [137] | attacker | corruption | budget | stated | none | · | explicit | conditional | stated |
| [146] | attacker | corruption | budget | stated | none | perturbation | explicit | universal | stated |
| [72] | learner | time | poly | n/a | corruption | · | explicit | existential | stated |
| [70] | learner | samples | class | stated | none | algo class | explicit | conditional | stated |
| [190] | learner | samples | class | stated | none | algo class | explicit | universal | stated |
| [22] | learner | queries | budget | stated | corruption | time | implicit | universal | stated |
| [58] | auditor | time | poly | stated | corruption | none | explicit | conditional | stated |
| [157] | none | none | unstated | n/a | none | · | · | existential | none |
| [152] | attacker | queries | budget | n/a | none | time | implicit | empirical | stated |
| [186] | none | none | unstated | n/a | none | none | · | existential | none |
| [130] | auditor | time | poly | stated | perturbation | none | explicit | existential | stated |
| [109] | auditor | time | poly | absent | corruption | samples | explicit | conditional | stated |
| [105] | auditor | time | poly | stated | none | none | explicit | existential | stated |
| [12] | none | none | unstated | n/a | none | none | · | empirical | none |
| [68] | auditor | time | poly | n/a | none | none | explicit | empirical | stated |
| [160] | attacker | corruption | measured | stated | none | · | explicit | empirical | stated |
| [129] | none | none | unstated | n/a | none | none | · | empirical | stated |
| [154] | none | none | unstated | n/a | none | none | · | empirical | stated |
| [133] | mechanism | memory | exponent | stated | none | · | explicit | existential | stated |
| [112] | learner | samples | exponent | stated | none | time | explicit | empirical | stated |
| [9] | none | none | unstated | n/a | none | none | · | conditional | stated |
| [45] | attacker | time | exponent | n/a | time | none | explicit | conditional | stated |
| [76] | learner | time | exponent | n/a | none | none | explicit | conditional | stated |
| [17] | learner | algo class | class | n/a | none | · | explicit | conditional | stated |
| [35] | auditor | algo class | class | n/a | none | none | explicit | universal | stated |
| [38] | attacker | circuit size | budget | n/a | circuit size | algo class | explicit | existential | stated |
| [110] | attacker | memory | exponent | n/a | memory | queries | explicit | universal | stated |
| [21] | attacker | algo class | class | n/a | none | algo class | explicit | conditional | stated |
| [44] | learner | time | conditional | n/a | none | algo class | explicit | conditional | stated |
| [24] | attacker | memory | exponent | n/a | memory | time | explicit | conditional | stated |
| [60] | attacker | algo class | class | stated | none | · | explicit | existential | stated |
| [118] | attacker | time | poly | n/a | none | none | explicit | conditional | stated |
| [148] | attacker | time | poly | n/a | none | none | explicit | conditional | stated |

*Table 6, continued*

| # | Party | Unit | Tightness | Baseline | 2nd party | 2nd notion | Assessed | Quantifier | Antecedent |
|---|---|---|---|---|---|---|---|---|---|
| [113] | learner | flops | exponent | stated | none | parameters | explicit | empirical | stated |
| [30] | none | none | unstated | n/a | none | none | · | empirical | stated |
| [147] | attacker | time | poly | stated | none | none | explicit | existential | stated |
| [155] | learner | flops | exponent | absent | none | parameters | explicit | conditional | stated |
| [62] | attacker | time | poly | stated | none | · | explicit | conditional | stated |
| [1] | attacker | circuit size | quasi-poly | stated | none | none | explicit | conditional | stated |
| [165] | learner | time | measured | stated | none | parameters | explicit | conditional | stated |
| [4] | mechanism | privacy budget | budget | n/a | none | none | explicit | empirical | stated |
| [37] | learner | time | poly | n/a | none | · | explicit | conditional | stated |
| [69] | learner | time | conditional | stated | none | algo class | explicit | conditional | stated |
| [47] | learner | time | exponent | stated | none | parameters | explicit | empirical | stated |
| [14] | none | none | unstated | n/a | none | none | · | empirical | none |
| [87] | mechanism | memory | conditional | stated | time | none | explicit | conditional | stated |
| [161] | mechanism | time | budget | stated | none | · | explicit | universal | stated |
| [18] | learner | flops | exponent | absent | none | parameters | explicit | conditional | stated |
| [29] | learner | samples | exponent | absent | none | parameters | explicit | conditional | stated |
| [19] | learner | time | measured | stated | none | none | explicit | conditional | stated |
| [94] | learner | algo class | class | stated | none | queries | explicit | universal | stated |
| [89] | attacker | algo class | class | stated | algo class | none | explicit | conditional | stated |
| [36] | learner | time | conditional | n/a | none | · | explicit | conditional | stated |
| [15] | learner | time | conditional | n/a | none | none | explicit | conditional | stated |
| [5] | attacker | queries | exponent | stated | queries | none | explicit | existential | stated |
| [26] | attacker | time | exponent | stated | time | none | explicit | conditional | stated |
| [23] | attacker | circuit size | exponent | n/a | none | none | explicit | existential | stated |
| [11] | attacker | time | conditional | n/a | none | none | explicit | conditional | stated |
| [167] | none | none | unstated | n/a | none | none | · | empirical | stated |
| [128] | learner | samples | exponent | stated | none | time | explicit | universal | stated |
| [103] | learner | circuit size | conditional | n/a | none | none | explicit | universal | stated |
| [104] | learner | algo class | class | n/a | none | · | explicit | universal | stated |
| [82] | learner | samples | exponent | stated | none | algo class | explicit | conditional | stated |
| [75] | learner | samples | exponent | absent | none | algo class | explicit | conditional | stated |
| [141] | learner | time | poly | stated | none | · | explicit | conditional | stated |
| [178] | learner | time | poly | stated | perturbation | · | explicit | conditional | stated |
| [31] | learner | time | measured | stated | none | samples | explicit | empirical | stated |
| [120] | none | none | unstated | n/a | none | · | · | universal | stated |
| [123] | learner | time | poly | n/a | none | algo class | explicit | existential | stated |
| [194] | mechanism | algo class | class | stated | none | time | implicit | conditional | stated |
| [8] | learner | algo class | class | n/a | none | samples | explicit | universal | stated |
| [77] | auditor | algo class | class | n/a | privacy budget | samples | explicit | existential | stated |
| [39] | none | none | unstated | n/a | none | none | · | conditional | stated |
| [91] | learner | algo class | class | n/a | none | none | explicit | universal | stated |
| [169] | none | none | unstated | n/a | none | none | · | universal | stated |
| [3] | learner | algo class | class | n/a | none | time | explicit | conditional | stated |
| [67] | learner | algo class | class | n/a | none | queries | explicit | universal | stated |
| [119] | learner | time | poly | n/a | none | none | explicit | conditional | stated |
| [183] | learner | samples | exponent | stated | none | none | explicit | conditional | stated |
| [106] | attacker | time | poly | n/a | privacy budget | none | explicit | conditional | stated |
| [56] | learner | algo class | class | n/a | none | none | explicit | universal | stated |
| [175] | learner | algo class | class | n/a | none | none | explicit | universal | stated |
| [191] | learner | algo class | class | n/a | none | · | explicit | conditional | stated |
| [64] | none | none | unstated | n/a | none | none | · | conditional | stated |
| [2] | learner | algo class | class | n/a | none | samples | explicit | existential | stated |
| [42] | attacker | queries | exponent | n/a | queries | none | explicit | universal | stated |
| [189] | attacker | algo class | class | stated | algo class | none | explicit | conditional | stated |
| [73] | learner | samples | exponent | stated | none | algo class | explicit | conditional | stated |
| [102] | learner | parameters | budget | stated | none | time | explicit | existential | stated |
| [33] | attacker | time | exponent | n/a | time | none | explicit | conditional | stated |
| [138] | attacker | time | exponent | n/a | none | none | explicit | conditional | stated |
| [48] | attacker | circuit size | quasi-poly | stated | none | none | explicit | conditional | stated |

*Table 6, continued*

| # | Party | Unit | Tightness | Baseline | 2nd party | 2nd notion | Assessed | Quantifier | Antecedent |
|---|---|---|---|---|---|---|---|---|---|
| [99] | attacker | circuit size | quasi-poly | stated | none | none | explicit | conditional | stated |
| [135] | learner | time | exponent | n/a | none | success prob | explicit | conditional | stated |
| [34] | learner | time | quasi-poly | n/a | none | algo class | explicit | existential | stated |
| [162] | learner | memory | budget | n/a | none | none | explicit | universal | stated |
| [20] | learner | time | conditional | stated | corruption | algo class | explicit | conditional | stated |
| [7] | learner | samples | exponent | stated | none | algo class | explicit | universal | stated |
| [88] | learner | samples | exponent | stated | none | time | explicit | universal | stated |
| [116] | auditor | algo class | class | n/a | none | · | explicit | conditional | stated |
| [132] | learner | flops | exponent | absent | none | parameters | explicit | empirical | stated |
| [27] | mechanism | time | poly | n/a | time | none | explicit | conditional | stated |
| [107] | attacker | circuit depth | exponent | n/a | none | queries | explicit | existential | stated |
| [139] | learner | samples | exponent | stated | none | algo class | explicit | universal | stated |
| [111] | learner | samples | exponent | stated | none | algo class | explicit | universal | stated |
| [125] | learner | samples | exponent | stated | none | time | explicit | conditional | stated |
| [81] | attacker | time | poly | n/a | privacy budget | none | explicit | universal | stated |
| [83] | attacker | memory | exponent | stated | memory | · | explicit | universal | stated |
| [115] | learner | parameters | budget | stated | none | samples | explicit | universal | stated |
| [46] | learner | parameters | budget | n/a | none | none | explicit | conditional | stated |
| [131] | learner | algo class | class | n/a | none | none | explicit | universal | stated |
| [198] | learner | time | poly | stated | none | samples | explicit | existential | stated |
| [41] | none | none | unstated | n/a | none | none | · | empirical | stated |
| [163] | mechanism | privacy budget | budget | stated | none | none | implicit | universal | stated |
| [85] | none | none | unstated | n/a | none | none | · | empirical | stated |
| [164] | attacker | samples | measured | stated | none | none | implicit | empirical | stated |

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